

2023 Higher Level Mathematics Paper 1

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Question 1

- (a) We know that $m \in \mathbb{R}$ which means it can be any number along the number line that satisfies $|5 + 3m| = 11$. The vertical lines refer to the *absolute value*, which means positive and negative numbers are all converted to positive.

Therefore we have:

$$5 + 3m = 11$$

$$3m = 11 - 5$$

$$3m = 6$$

$$\implies m = 2$$

but remember, absolute value converts negative to positive, so we also have:

$$5 + 3m = -11$$

$$3m = -11 - 5$$

$$3m = -16$$

$$\implies m = -\frac{16}{3}$$

and both of these solutions are on $\mathbb{R}$, the real number line, so we have $m = 2$ **and** $m = -\frac{16}{3}$.